

Практика - Брокер сообщений Apache Kafka

Выполнил - Гусев Кирилл

Группа - Управление на основе данных

1 Упражнение

Задача 1: Запуск Apache Kafka

```
[+] Running 15/15
✓ kafka1 11 layers [██████████] 0B/0B Pulled 97.8s
✓ 3550ee360766 Pull complete 8.2s
✓ e3455c6406eb Pull complete 42.5s
✓ 95cdaa6975d4 Pull complete 3.6s
✓ 3d3883ed71ab Pull complete 4.6s
✓ c5df82bd3053 Pull complete 19.0s
✓ 40036816fad0 Pull complete 34.7s
✓ f7e8fecc2b94 Pull complete 19.8s
✓ e80d13bd6d87 Pull complete 21.1s
✓ a11e73b0229d Pull complete 22.2s
✓ be6fe5229c1d Pull complete 60.6s
✓ 01424b4d5bdc Pull complete 43.5s
✓ zoo1 2 layers [██] 0B/0B Pulled 97.9s
✓ d23edc08b858 Pull complete 60.0s
✓ 9b1cb94f7c7c Pull complete 35.9s
[+] Running 3/3
✓ Network kafka_default Created 0.7s
✓ Container zoo1 Started 1.4s
✓ Container kafka1 Started 0.5s
```

Задача 2. Создание темы в Kafka

```
[appuser@kafka1 ~]$ kafka-topics --create --topic logs_text --partitions 1 --replication-factor 1 --bootstrap-server localhost:9092
WARNING: Due to limitations in metric names, topics with a period ('.') or under score ('_') could collide. To avoid issues it is best to use either, but not both.
Created topic logs_text.
[appuser@kafka1 ~]$
```

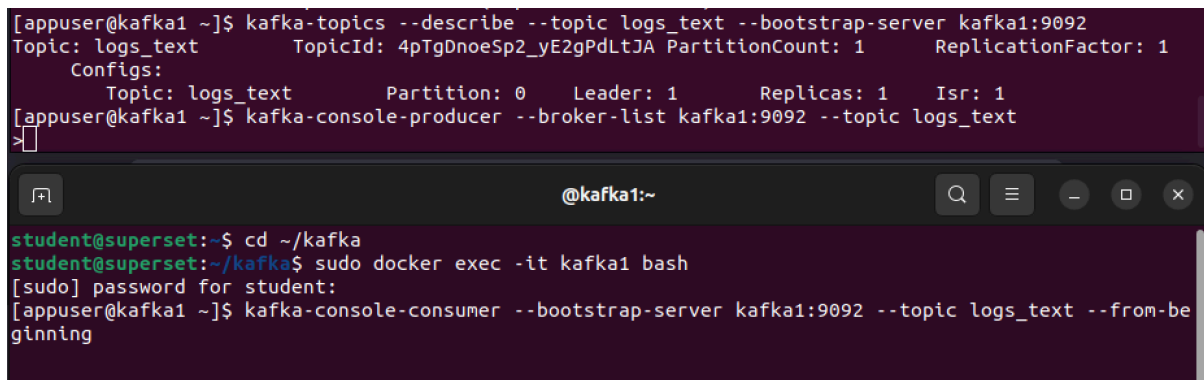
```
[appuser@kafka1 ~]$ kafka-topics --describe --topic logs_text --bootstrap-server kafka1:9092
Topic: logs_text      TopicId: 4pTgDnoeSp2_yE2gPdLtJA PartitionCount: 1      R
eplicationFactor: 1   Configs:
      Topic: logs_text Partition: 0    Leader: 1      Replicas: 1      I
sr: 1
[appuser@kafka1 ~]$
```

После просмотра содержимого logs_text видно, что он содержит:

PartitionCount: 1 – одна партиция
ReplicationFactor: 1 – одна реплика
Leader: 1 – лидер брокер 1
Isr: 1 – синхронизированные реплики

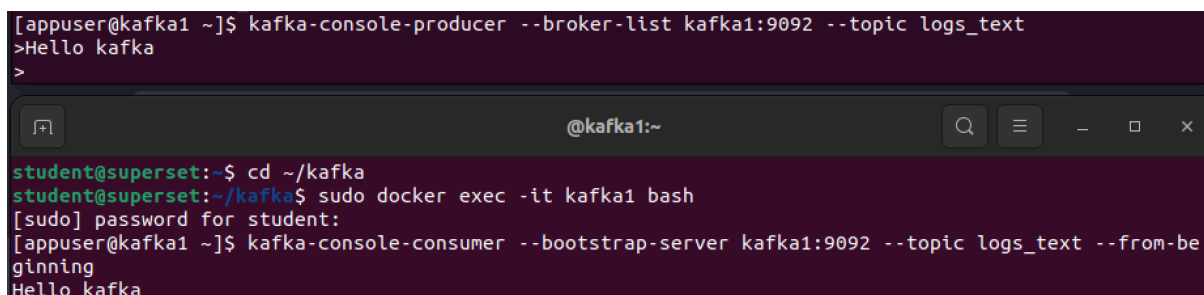
Задача 3. Создание консольных потребителей и продюсеров

```
[appuser@kafka1 ~]$ kafka-topics --describe --topic logs_text --bootstrap-server kafka1:9092
Topic: logs_text      TopicId: 4pTgDnoeSp2_yE2gPdLtJA PartitionCount: 1      ReplicationFactor: 1
Configs:
  Topic: logs_text    Partition: 0    Leader: 1      Replicas: 1    Isr: 1
[appuser@kafka1 ~]$ kafka-console-producer --broker-list kafka1:9092 --topic logs_text
>
```



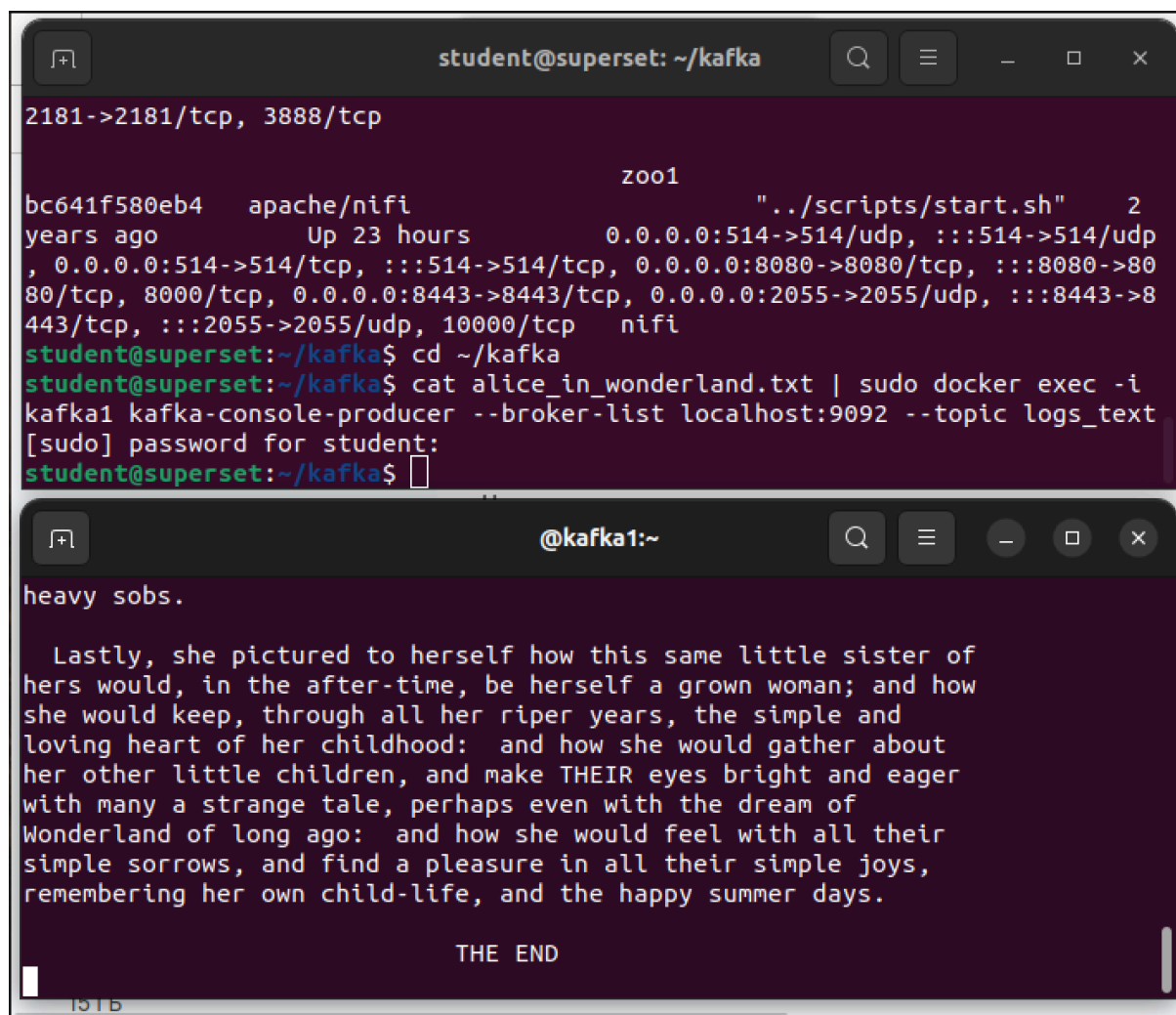
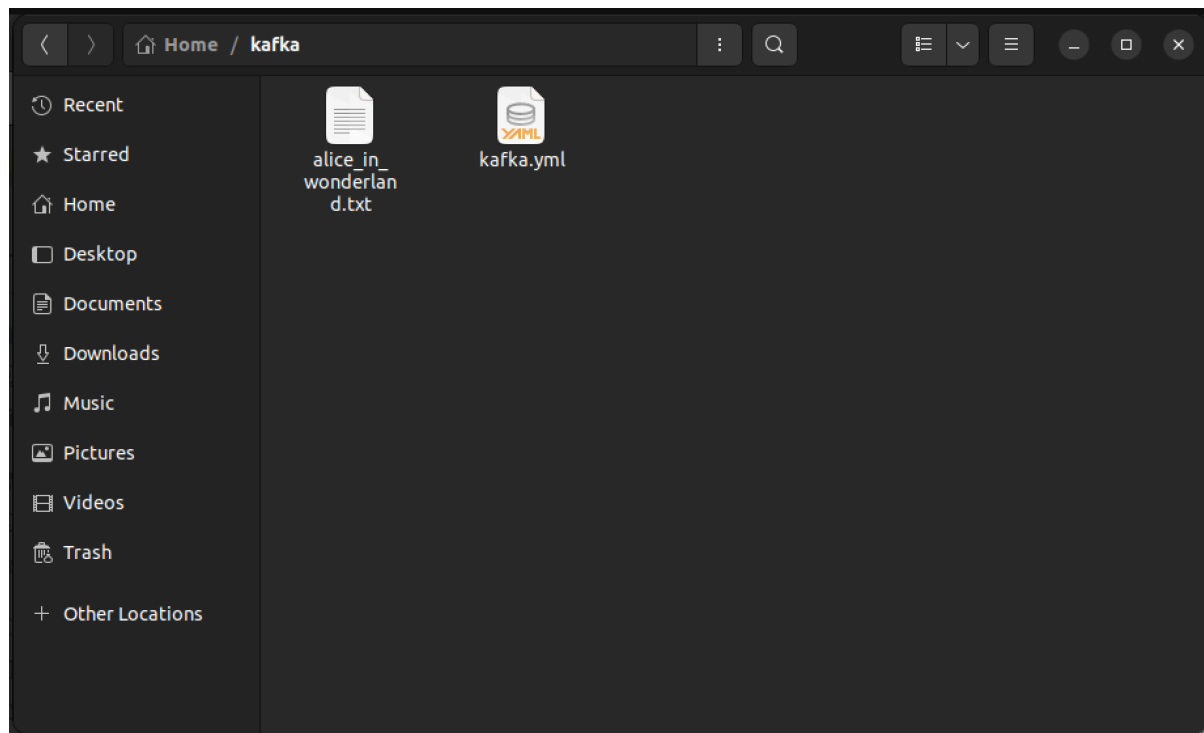
The screenshot shows a terminal window with a dark background. The first command executed is `kafka-topics --describe --topic logs_text --bootstrap-server kafka1:9092`, which displays the details of the `logs_text` topic. The second command is `kafka-console-producer --broker-list kafka1:9092 --topic logs_text`, which starts the producer console. The prompt `>` indicates that the producer is ready to accept input.

```
[appuser@kafka1 ~]$ kafka-console-producer --broker-list kafka1:9092 --topic logs_text
>Hello kafka
>
```



The screenshot shows a terminal window with a dark background. The first command executed is `kafka-console-producer --broker-list kafka1:9092 --topic logs_text`. The user enters `Hello kafka` and presses enter. The prompt `>` indicates that the producer is ready to accept input. The second command is `kafka-console-consumer --bootstrap-server kafka1:9092 --topic logs_text --from-beginning`, which starts the consumer console. The prompt `>` indicates that the consumer is ready to receive messages.

Задача 3.1 передача файла через продюсера и потребителя



Упражнение 2 - Flafka

1. Установим Flume

```
student@superset: ~/kafka
181.214.104, 2a01:4f9:3a:2c57::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443
... connected.
HTTP request sent, awaiting response... 200 OK
Length: 87380462 (83M) [application/x-gzip]
Saving to: 'apache-flume-1.11.0-bin.tar.gz.1'

apache-flume-1.11. 100%[=====] 83,33M 447KB/s in 2m 0s

2026-02-23 21:47:26 (712 KB/s) - 'apache-flume-1.11.0-bin.tar.gz.1' saved [8
7380462/87380462]

student@superset:~/kafka$
```

2. Запустили Flume

```
student@superset: ~/kafka/flume
student@superset:~/kafka/flume$ ^[[200~/bin/flume-ng agent \
> --conf ./conf \
> --conf-file ~/kafka/spooldir.conf \
> --name a1 \
> ^C
student@superset:~/kafka/flume$ ./bin/flume-ng agent --conf ./conf --conf-fi
le ~/kafka/spooldir.conf --name a1 -Dflume.root.logger=INFO,console
Info: Including Hive libraries found via () for Hive access
+ exec /usr/lib/jvm/java-11-openjdk-amd64/bin/java -Xmx20m -Dflume.root.logg
er=INFO,console -cp '/home/student/kafka/flume/conf:/home/student/kafka/flum
e/lib/*:/lib/*' -Djava.library.path= org.apache.flume.node.Application --con
f-file /home/student/kafka/spooldir.conf --name a1
```

3. Передали файл через конвейер данных Flafka

```
student@superset: ~/kafka/flume
> --name a1 \
> ^C
student@superset:~/kafka/flume$ ./bin/flume-ng agent --conf ./conf --conf-file
~/kafka/spooldir.conf --name a1 -Dflume.root.logger=INFO,console
Info: Including Hive libraries found via () for Hive access
+ exec /usr/lib/jvm/java-11-openjdk-amd64/bin/java -Xmx20m -Dflume.root.logg
er=INFO,console -cp '/home/student/kafka/flume/conf:/home/student/kafka/flum
e/lib/*:/lib/*' -Djava.library.path= org.apache.flume.node.Application --con
f-file /home/student/kafka/spooldir.conf --name a1
█

student@superset: ~/kafka
student@superset:~$ cd ~/kafka
student@superset:~/kafka$ cp alice_in_wonderland.txt ~/kafka/spool/
student@superset:~/kafka$ █

@kafka1:~
her other little children, and make THEIR eyes bright and eager
with many a strange tale, perhaps even with the dream of
Wonderland of long ago: and how she would feel with all their
simple sorrows, and find a pleasure in all their simple joys,
remembering her own child-life, and the happy summer days.

THE END
█
```